



A Formal Framework for Ontological Vagueness in Intelligence Analysis

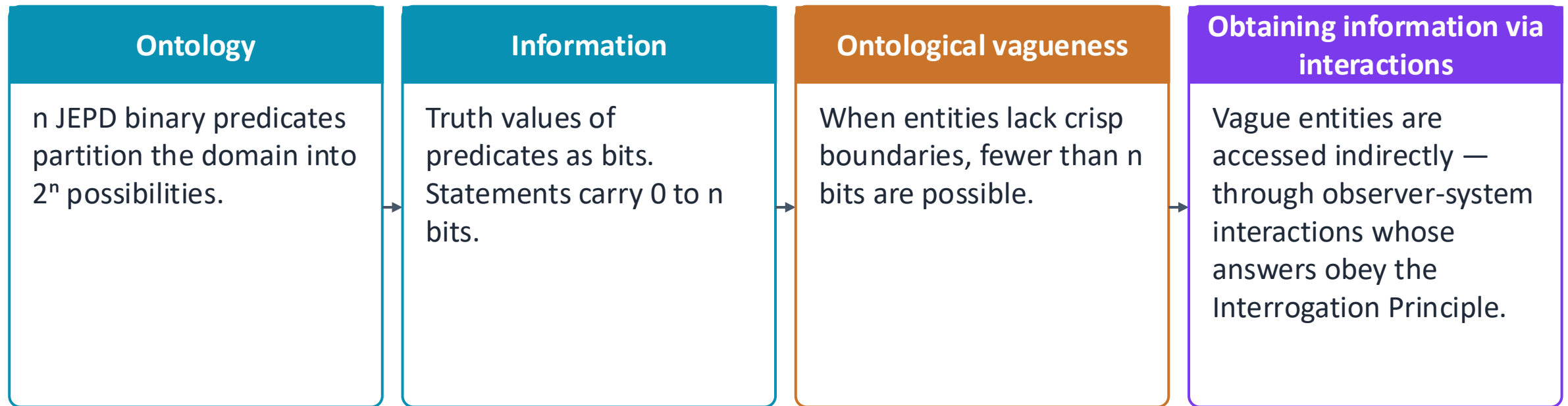
Towards Trustworthy and Explainable Judgment

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Four concepts to clarify

A clean chain from ontology to ontological vagueness, and how information about vague entities is obtained



Concepts 1 & 2 · Ontology + Information

JEPD binary predicates partition a domain; information capacity follows directly.

Ontology → information capacity

Specify a domain by n JEPD binary predicates $q_0, \dots, q_{n-1}$ (jointly exhaustive, pairwise disjoint). They partition the domain into **$N = 2^n$ possibilities**, giving **$icap = \log_2(N) = n$ bits** of capacity.

A true statement carries between 0 and icap bits — depending on how many of the N possibilities it excludes (each halving of the surviving set = +1 bit).

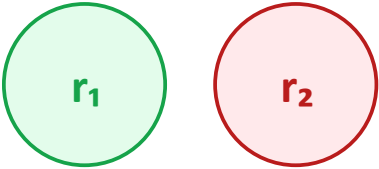
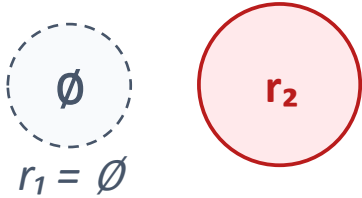
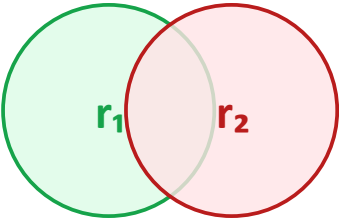
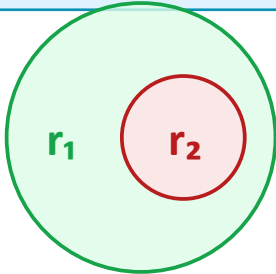
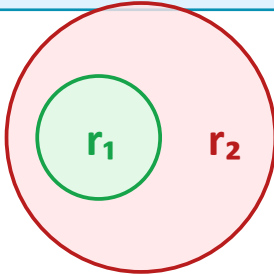
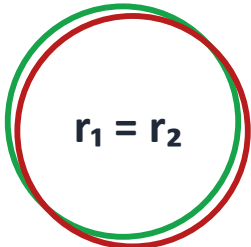
Three running examples — true statements at different bit-levels

Domain	N	n	icap	Example true statements (with bit content)	
1 sensor cell (HIT / MISS)	2	1	1 bit	0 bits	T (or $HIT(s) \vee MISS(s)$) — vacuous — excludes 0 of 2
				1 bit	$HIT(s)$ — atomic — excludes 1 of 2
2 sensor cells (HH / HM / MH / MM)	4	2	2 bits	1 bit	$HIT(s_1)$ — atomic — excludes 2 of 4
				1 bit	$(HIT(s_1) \wedge HIT(s_2)) \vee (MISS(s_1) \wedge MISS(s_2))$ "sensors agree" — correlational — excludes 2 of 4
				2 bits	$HIT(s_1) \wedge MISS(s_2)$ — atomic — excludes 3 of 4

Example · mereology (n = 3)

Region pictures of an ordered pair (OPAREA-Alpha, OPAREA-Bravo) and their q_0 q_1 q_2 truth-value triples

$q_0 = (r_1 \cap r_2 \text{ is non-empty})$ · $q_1 = (r_1 \subseteq r_2)$ · $q_2 = (r_2 \subseteq r_1)$ $N = 8$, $n = 3$, $icap = \log_2(N) = 3 \text{ bits}$

$q_0=0$ $q_1=0$ $q_2=0$ DR  <i>disjoint</i>	$q_0=0$ $q_1=0$ $q_2=1$ ⊥₀₁  <i>r₁ present, r₂ empty</i>	$q_0=0$ $q_1=1$ $q_2=0$ ⊥₁₀  <i>r₁ empty, r₂ present</i>	$q_0=0$ $q_1=1$ $q_2=1$ ⊥₁₁  <i>both empty</i>
$q_0=1$ $q_1=0$ $q_2=0$ PO  <i>partial overlap</i>	$q_0=1$ $q_1=0$ $q_2=1$ PPi  <i>r₂ proper part of r₁</i>	$q_0=1$ $q_1=1$ $q_2=0$ PP  <i>r₁ proper part of r₂</i>	$q_0=1$ $q_1=1$ $q_2=1$ EQ  <i>r₁ = r₂ (coincident)</i>

Concept 3 · Ontological vagueness

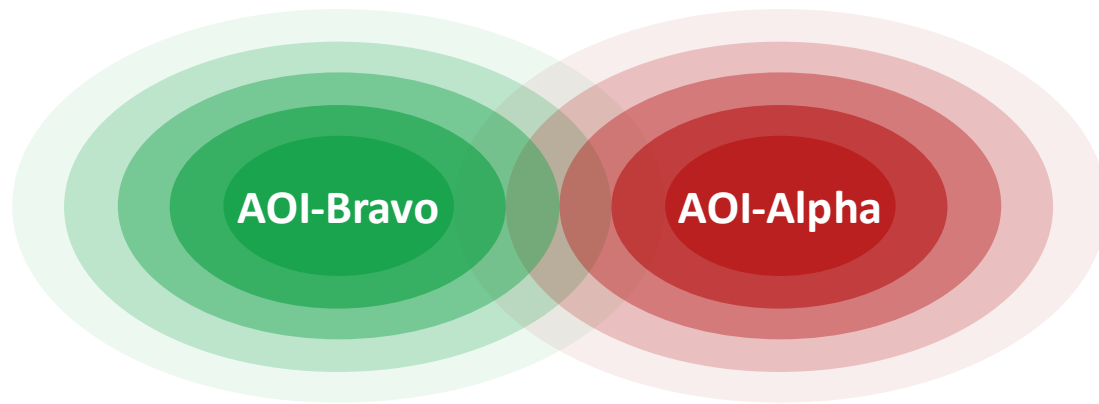
When entities have indeterminate boundaries / properties, fewer than n bits are possible

Information capacity is structurally limited

Crisp domain: 2^n configurations, up to n bits possible.

Vague domain: the same 2^n configurations are addressable in principle, but **fewer than n bits** are actually realisable. Some predicates lack a determinate truth value.

$n = 3$ · AOI-Alpha and AOI-Bravo (graded, overlapping boundaries)



Graded boundaries. The mixed band in the middle is the overlap of the two fuzzy zones.

AOI-Alpha and **AOI-Bravo** are real adversary forces operating in the same theater. Their areas of influence are not bounded by crisp lines but by **graded zones** — the contested zone belongs somewhat to each force, somewhat to neither.

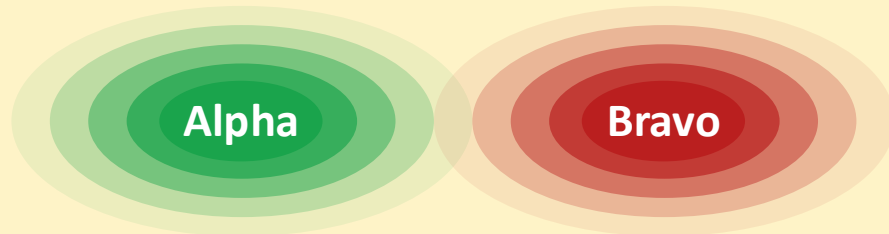
Where the graded zones overlap, the predicates q_0, q_1, q_2 do not all admit determinate truth values.

→ $iscap < 3 \text{ bits}$

Example · vague (AOI-Alpha, AOI-Bravo) and its istate

Vague pair $\rightarrow$ two crisp branches $\rightarrow$ istate $= (1/\sqrt{2}) |DR\rangle + (1/\sqrt{2}) |PO\rangle$

The vague configuration

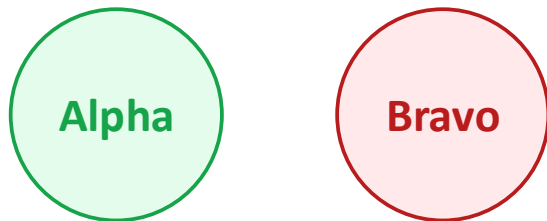


Predicate truth-value pattern

$q_0 = ?$ <i>(gappy)</i> $q_0 = \text{overlap} ?$	$q_1 = 0$ <i>(F)</i> $q_1 = \text{Alpha} \subseteq \text{Bravo} ?$	$q_2 = 0$ <i>(F)</i> $q_2 = \text{Bravo} \subseteq \text{Alpha} ?$
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Branch 1 · $q_0 = 0$

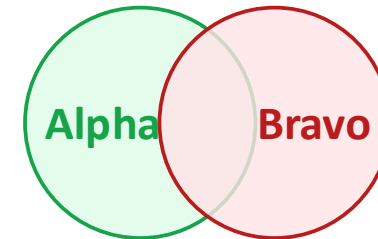
DR



$(1/\sqrt{2}) |DR\rangle$

Branch 2 · $q_0 = 1$

PO

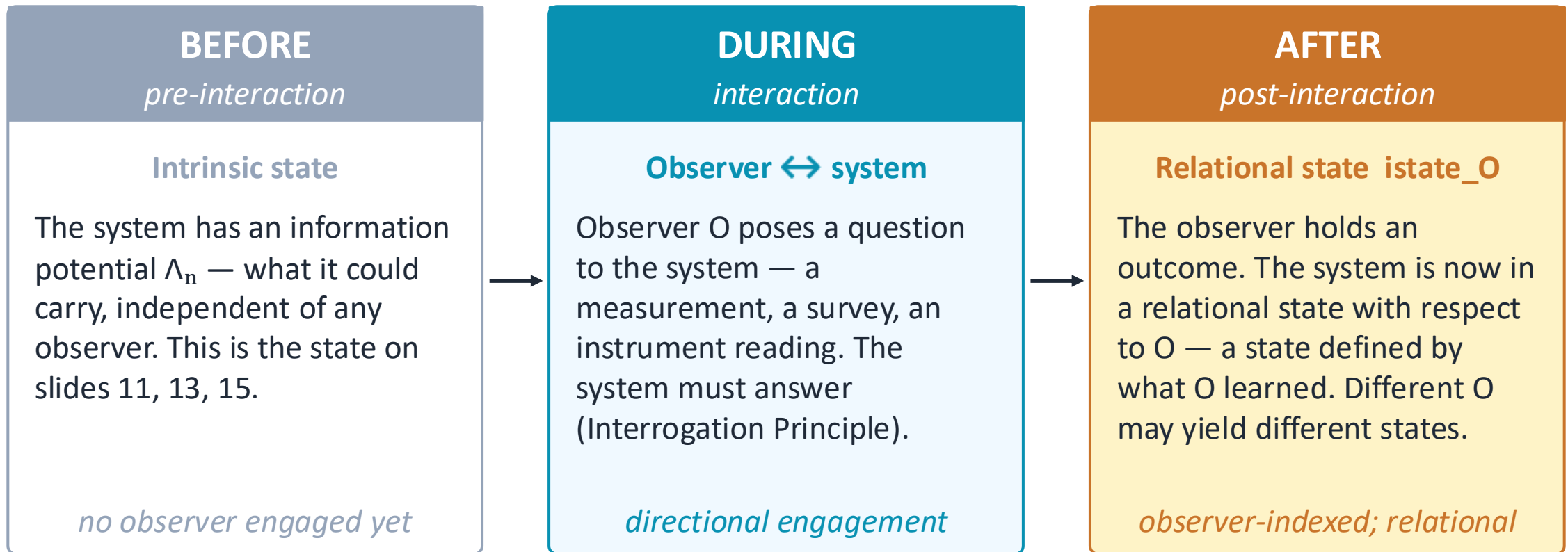


$(1/\sqrt{2}) |PO\rangle$

istate(AOI-Alpha, AOI-Bravo) $= (1/\sqrt{2}) |DR\rangle + (1/\sqrt{2}) |PO\rangle$. Two of 8 RCC5⁺ basis vectors non-zero $\rightarrow$ **icap = 2 bits** < 3 bits.

Concept 4 · What is an interaction?

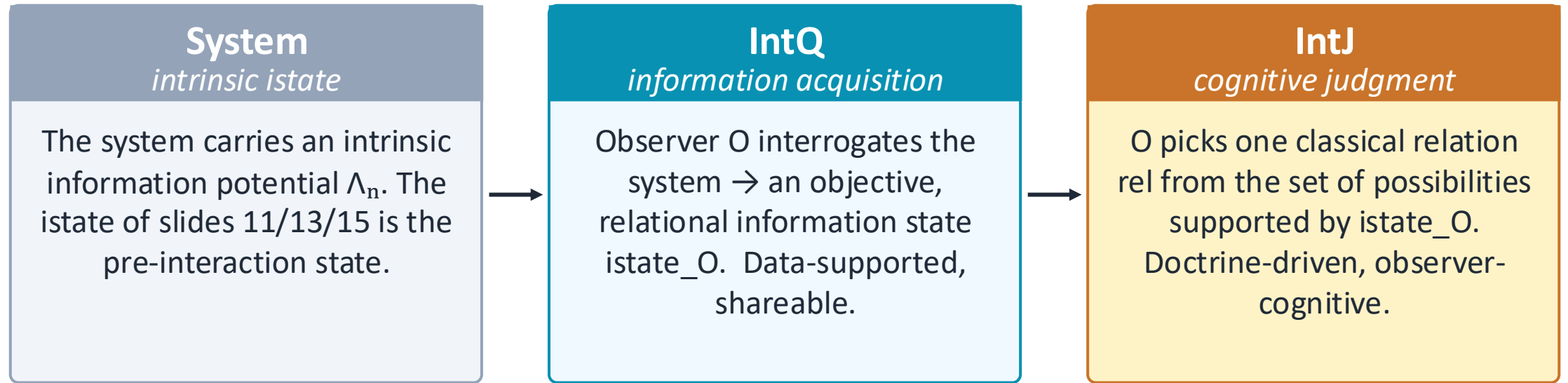
An observer engaging with a system to obtain information. Changes both sides — observer learns; system is left in a relational state.



Vocabulary used in the next slides: *pre-interaction* (the intrinsic potential) vs *post-interaction $istate_O$* (the relational state with respect to a specific observer O).

Two-stage interaction: IntQ (acquire) → IntJ (judge)

Every analyst-system interaction has two distinct stages — record both for auditability.



Auditable chain: **IntQ** is data-supported and relational; **IntJ** is doctrine-driven and cognitive. Recording both stages lets failures localise to one or the other.

The Interrogation Principle: systems are forced to answer

The system must answer every question — capacity-disrespecting queries force a random determinate answer.

Disrespecting Q on (AOI-Bravo, AOI-Alpha) · (G, F, F),
icap = 2

Hypothesised istate (from slide 18):

$$|(\text{AOI-Bravo}, \text{AOI-Alpha})\rangle = (1/\sqrt{2}) |DR\rangle + (1/\sqrt{2}) |PO\rangle$$

$Q_0? \odot Q_1? \odot Q_2?$ (3 bits > icap = 2 →
disrespecting)

Forced answer (Born rule, no preferred basis
vector):

DR (F, F, F) $p = 1/8$	PO (T, F, F) $p = 1/8$	PP (T, T, F) $p = 1/8$	PPi (T, F, T) $p = 1/8$
EQ (T, T, T) $p = 1/8$	⊥₀₁ (F, F, T) $p = 1/8$	⊥₁₀ (F, T, F) $p = 1/8$	⊥₁₁ (F, T, T) $p = 1/8$

*The 3-bit question exceeds icap = 2 — no preferred
answer. Born rule: every RCC5⁺ basis vector at $p =$
1/8.*

The Interrogation Principle

*"An elementary system must answer every question posed to it. If an interaction demands more information than the system's intrinsic capacity, the system must still provide an answer by **randomly** 'crispening' its state with respect to the observer."*

Consequence:

Even on a gappy predicate, the system is forced to produce a determinate Y/N answer when asked. The randomness is the operational signature of intrinsic vagueness.

"DR" is not "DR" is not "DR": same answer, three meanings

Three different questions can all yield the judgment $DR(AOI\text{-}Bravo, AOI\text{-}Alpha)$. The epistemic content of each DR is very different.

Setup: (AOI-Bravo, AOI-Alpha) is a vague pair with intrinsic $icap = 2$ (q_0 gappy, $q_1 = q_2 = N$). Three different questions all produce DR — by very different mechanisms.

(i) 3-bit question

capacity-disrespecting · exceeds $icap = 2$

$Q_0 \odot Q_1 \odot Q_2$

"Do they overlap, is AOI-Bravo part of AOI-Alpha, is AOI-Alpha part of AOI-Bravo?"

Path to DR

3-bit question on a 2-bit system — *no preferred answer*.

Born rule selects **1 of 8** RCC5⁺ basis vectors uniformly ($\frac{1}{8}$ each), then $\perp_{01}$, $\perp_{10}$, $\perp_{11}$ **redistribute** uniformly across the 5 RCC5 judgments.

DR : $\frac{1}{8} + 3 \cdot (1/40) = 1/5$. Every RCC5 judgment gets $1/5$.

$$P(DR) = 0.20$$

DR FROM EXCEEDED CAPACITY

uniform over all 5 RCC5 · no signal

(ii) 2-bit FGF

wrong hypothesis · treats q_1 as gappy

$Q_0 \odot Q_2$

"Do they overlap? Is AOI-Alpha part of AOI-Bravo?" (q_1 left unasked.)

Path to DR

Predicted istate $\{DR, \perp_{10}\}$.

$\perp$ branches carry *no mereological information* and **redistribute uniformly** over the 5 RCC5 judgments (foise).

DR : $6/10$, PO, PP, PPI, EQ : $1/10$ each.

$$P(DR) = 0.60$$

DR FROM A LEAKY ANSWER

DR-leaning bias from the $\perp$ -redistribution

(iii) 2-bit GFF

right hypothesis · treats q_0 as gappy

$Q_1 \odot Q_2$

"Is AOI-Bravo part of AOI-Alpha? Is AOI-Alpha part of AOI-Bravo?" (q_0 left unasked.)

Path to DR

Predicted istate $\{DR, PO\}$.

No $\perp$ branch — *both alternatives carry full mereological content*.

Impartial IntJ picks **DR or PO** by Born rule.

System left in its ground state — no disturbance.

$$P(DR) = \frac{1}{2}$$

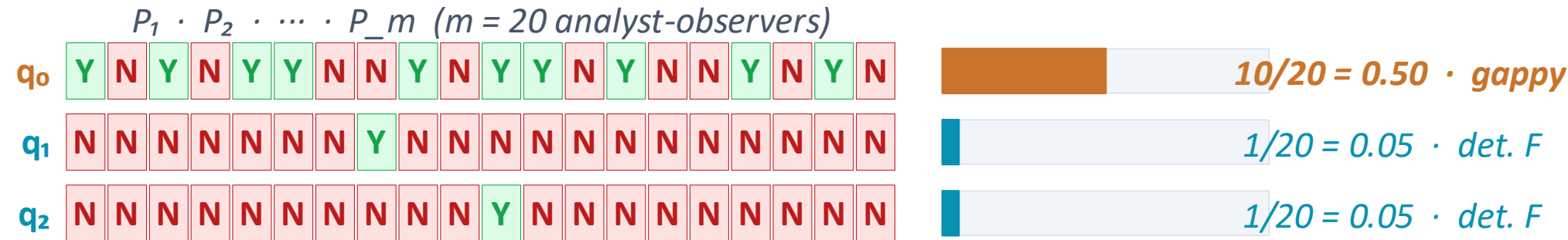
DR FROM AN HONEST CHOICE

between two genuine alternatives

Take-away: the classical judgment DR alone does not reveal which mechanism produced it. *Auditable reporting must carry the question pattern alongside the judgment* — a 3-bit DR, a FGF DR, and a GFF DR are not interchangeable evidence.

Survey → frequencies → istate

m analyst-observers fill an $n \times m$ matrix; squared istate amplitudes equal observed frequencies (Born-rule analogue).



Born-rule analogue

$$|\alpha_k|^2 = f_k \quad (\text{so } \alpha_k = \sqrt{f_k} \text{ for each } k \in \{DR, PO, PP, PPI, EQ, \perp_{01}, \perp_{10}, \perp_{11}\})$$

α_k = istate amplitude on basis vector k · f_k = relative frequency m_k / m with which analyst-observers judged k

Recovered istate

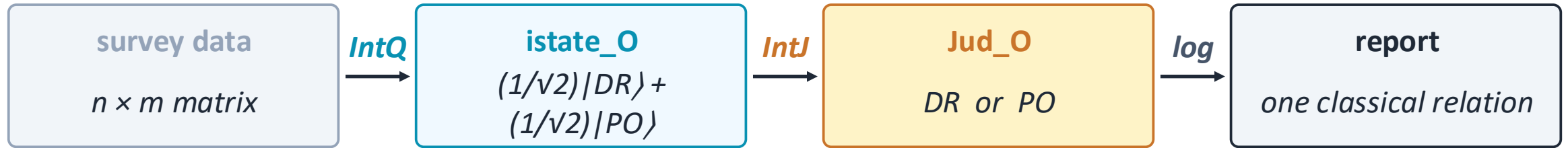
$$f_{\{DR\}} = 0.5, f_{\{PO\}} = 0.5 \rightarrow \sqrt{} \rightarrow \alpha_{\{DR\}} = 1/\sqrt{2}, \alpha_{\{PO\}} = 1/\sqrt{2}$$

$$\text{istate}_O(\text{AOI-Alpha}, \text{AOI-Bravo}) = (1/\sqrt{2}) |DR\rangle + (1/\sqrt{2}) |PO\rangle$$

Subscript O records that this is O's istate — built from O's sample of analyst-observers.

Analyst's IntJ and the auditable chain

With *istate_O* established, *O* picks one relation for the report. The full chain is auditable.



IntJ — two doctrine paths from the same *istate_O*:

Born-rule baseline

No doctrine — pick randomly from $\{DR, PO\}$ with weights $|1/\sqrt{2}|^2 = 50\%$ each. Honest under uncertainty.

$$\text{Jud}_O \in \{DR, PO\}$$

Doctrine-driven (thick judger)

Apply expert heuristic — "neighbouring regions overlap $\rightarrow PO$ " or "state boundaries divide $\rightarrow DR$ ". Defensible if doctrine is independently warranted.

$$\text{Jud}_O = PO \text{ (or DR)}$$

Auditability: locating failures

If the report is wrong, identify which stage failed: **IntQ fault** $\rightarrow$ bad survey design, biased / unrepresentative population, wrong question set. **IntJ fault** $\rightarrow$ valid *istate*, but the analyst's doctrine selected a relation poorly supported by other evidence.

The framework's contribution is this decomposition — without it, the report is a black box.

Summary · three vague mereological examples

Same $n = 3$ budget; different determinacy patterns reduce *icap* from 3 bits to 2 bits

Concept	(AOI-Alpha, AOI-Bravo) G-mode	(observed activity zone, suspected operational area) G + classical correlation	(R_1, R_2) ecoregion purely C-mode
Ontology <i>predicates · regions</i>	$n = 3 \cdot q_0, q_1, q_2$ regions: AOI-Alpha, AOI-Bravo	$n = 3 \cdot q_0, q_1, q_2$ regions: observed zone OBS, suspected area SUS	$n = 3 \cdot q_0, q_1, q_2$ contested sectors (R_1, R_2) over campaign-phase labels A, B
Information <i>max budget</i>	max 3 bits	max 3 bits	max 3 bits
Vagueness <i>bits + determinacy pattern</i>	icap = 2 bits · istate = { DR, PO } pattern: $q_0 \text{ G } \cdot q_1 = \text{F D} \cdot q_2 = \text{F D}$ mode: G (no q -correlation)	icap = 2 bits · istate = { PP, EQ } pattern: $q_0 = \text{T D} \cdot q_1 = \text{T D} \cdot q_2 \text{ G}$ mode: G + classical correlation $q_0 \leftrightarrow q_1$	icap = 2 bits · istate = { DR, EQ } pattern: $q_0 \text{ G } \cdot q_1 \text{ G } \cdot q_2 \text{ G}$ mode: C + joint correlation $q_0 = q_1 = q_2$
Hilbert space <i>ket-vector form</i>	$ \psi\rangle = (1/\sqrt{2}) DR\rangle + (1/\sqrt{2}) PO\rangle$ spans { $ 000\rangle, 100\rangle$ } $\subset \mathcal{H}$	$ \psi\rangle = (1/\sqrt{2}) PP\rangle + (1/\sqrt{2}) EQ\rangle$ spans { $ 110\rangle, 111\rangle$ } $\subset \mathcal{H}$	$ \psi\rangle = (1/\sqrt{2}) DR\rangle + (1/\sqrt{2}) EQ\rangle$ spans { $ 000\rangle, 111\rangle$ } $\subset \mathcal{H}$ (GHZ-style)

All three give **icap = 2 bits < 3 bits**, but for structurally different reasons. **G** = individual gaps; **C** = joint correlation across gappy individuals (no classical analogue).

The three modes · D / G / C

Three algebraic modes of indeterminacy

D

Determinate
(classical fragment)

Every constituent entity has a unique determinate for each predicate (Wilson 2013). q_i is pinned TRUE or FALSE. Crisp boundaries, sharp parthood — the regime classical mereology already handles.

example

single-sensor target detected: HIT(s) pinned TRUE · n bits available

G

Gappy
(individual indeterminacy)

q_i lacks a determinate truth value because at least one constituent entity possesses the relevant determinable without a unique determinate (Wilson 2013). Each region's truth-value is independently absent — no q-correlation.

example

(AOI-Alpha, AOI-Bravo): q_0 gappy · $q_1 = q_2 = F$ · no q-correlation

C

Correlated
(entangled indeterminacy)

Each q_i is gappy in the Wilson sense (determinable without unique determinate), yet a joint correlation across the q_i is fully determinate. Indeterminacy that travels in lockstep across regions — the novel ontological category.

example

contested-sector classification (R_1, R_2): q_0, q_1, q_2 all gappy; the three always agree (DR or EQ)

Every vague configuration is a mix of D / G / C across its predicates. C-mode is the genuinely novel category absent from classical mereology.

Stabilizer formalism as ontological classifier

The stabilizer formalism — the algebraic backbone of quantum error correction — is repurposed as a classification tool for vague spatial entities. Only the real (rebit) fragment is invoked; every computation runs on classical hardware.

Generators encode mereological information:

- Z-type** — information present (determinate)
- X-type** — information absent (ontological indeterminacy)

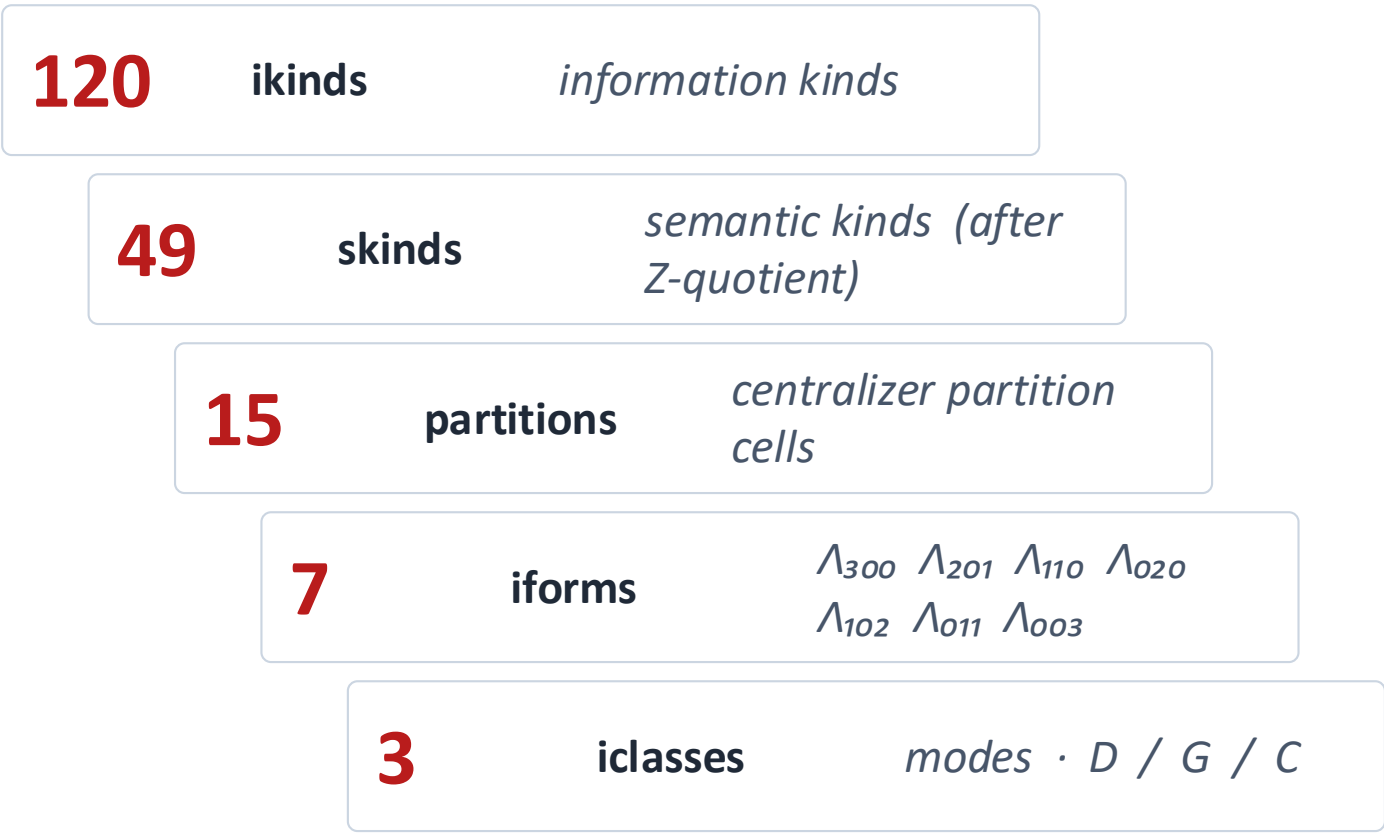
REBIT FRAGMENT
ONLY
Classical hardware suffices.
No quantum computer required.

WORKED EXAMPLE · (AOI-Bravo, AOI-Alpha)
predicates · q_0 overlap = GAPPY · q_1 parthood = FALSE · q_2 inverse parthood = FALSE

ikind	$\langle +X_0, +Z_1, +Z_2 \rangle$	<i>labelled stabilizer group · Z generators carry signs</i>
skind	$[X_0, +Z_1, +Z_2]$	<i>phase abstracted on X; Z-signs preserved</i>
partition	$\{ \text{TTD}, \text{TFG}, \text{FTG}, \text{FFG} \}$	<i>centralizer partition cell of (AOI-Bravo, AOI-Alpha)</i>
iform	$\{ \text{DDG}, \text{DGD}, \text{GDD} \}$	<i>position-equivalence class · 2 D-bits · 1 G-predicate</i>
modes	$D / G / C$	<i>the three iclasses; $(\text{AOI-Bravo}, \text{AOI-Alpha}) \in \{D, G\}$</i>

Closed-world classification hierarchy

Each level is a quotient of the level above by a known equivalence. Every candidate system is one of the 49 skinds (or 120 i kinds).



Quotients

- 120 / Z-equivalence → 49
- 49 / centralizer → 15
- 15 / iform shape → 7
- 7 / mode signature → 3

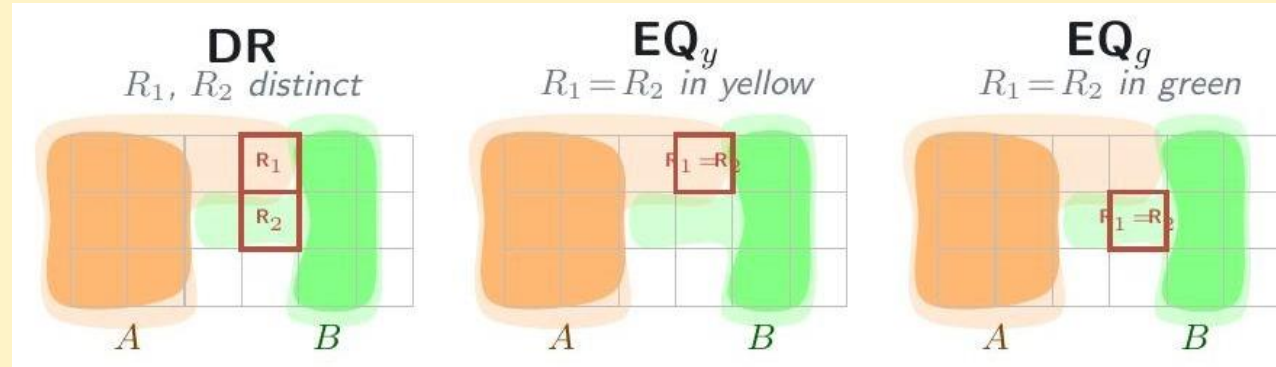
Identifying the type of an unknown system is a **closed-world classification problem**: every candidate is one of the 49 skinds (or 120 i kinds).

(AOI-Bravo, AOI-Alpha) at each level: ikind { +X₀, +Z₁, +Z₂ } → skind [X₀, +Z₁, +Z₂] → partition { TTD, TFG, FTG, FFG } → iform { DDG, DGD, GDD } → **mode D / G / C**

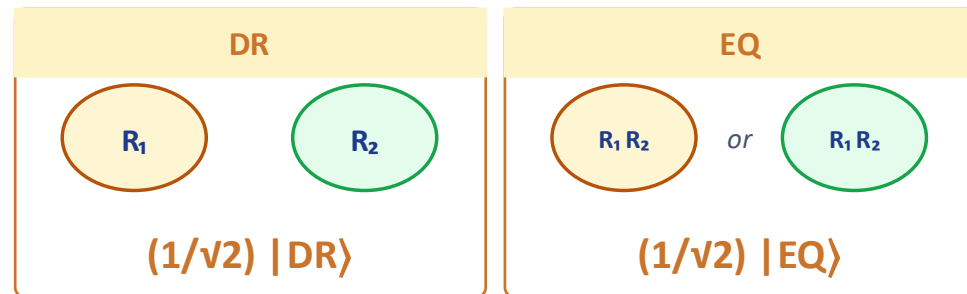
Example · contested sectors (R_1, R_2)

Each crisp RCC5 relation is a basis vector; possible = non-zero coefficient, ruled out = coefficient 0

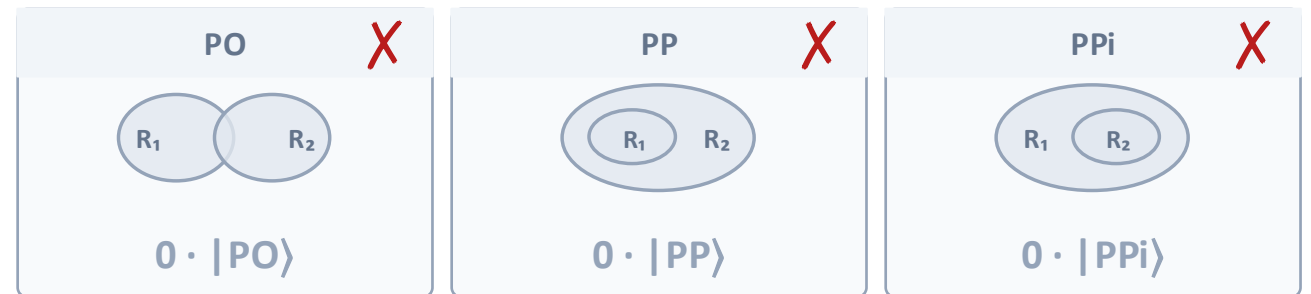
The vague configuration



Possible crisp configurations (non-zero coefficient)



Ruled out (coefficient = 0)



$$\text{istate}(R_1, R_2) = |(R_1, R_2)\rangle = (1/\sqrt{2}) |DR\rangle + (1/\sqrt{2}) |EQ\rangle + 0 \cdot |PO\rangle + 0 \cdot |PP\rangle + 0 \cdot |PPi\rangle + 0 \cdot |\perp_{01}\rangle + 0 \cdot |\perp_{10}\rangle + 0 \cdot |\perp_{11}\rangle$$

Normalisation: $|1/\sqrt{2}|^2 + |1/\sqrt{2}|^2 = 1$. 2 of 8 basis vectors carry non-zero coefficient $\rightarrow$ icap = 2 bits < 3 bits.

Formal istate of (R_1, R_2)

$q_0 = q_1 = q_2$ jointly pinned, individually gappy $\leftrightarrow$ $istate = \{ DR, EQ \}$

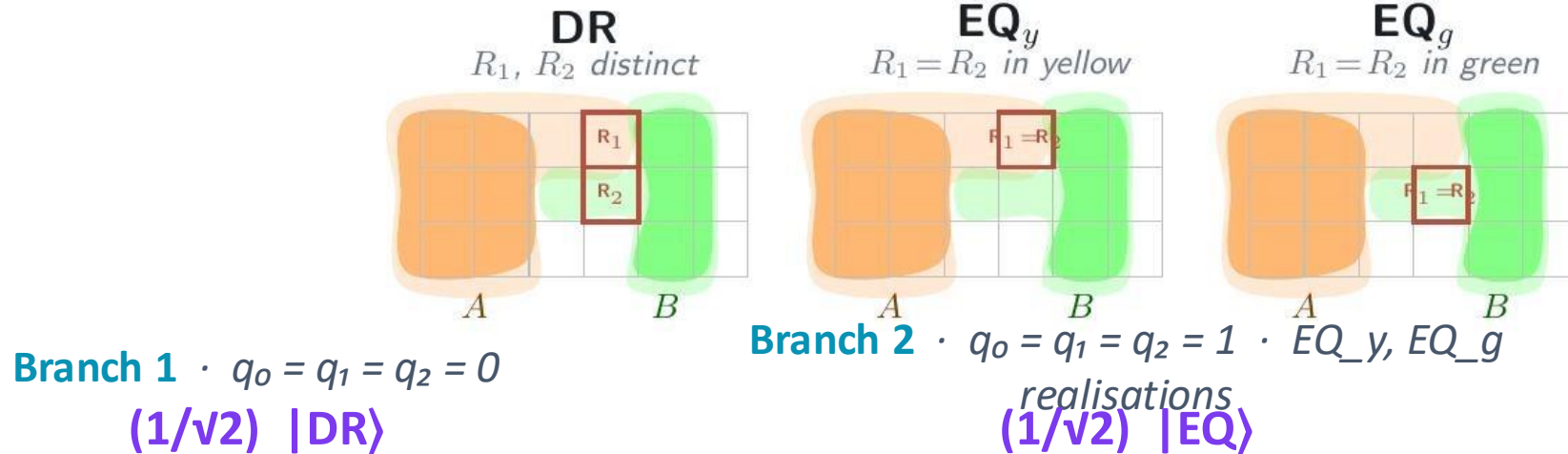
Predicate truth-value pattern

$q_0 = ?$ (gappy / indeterminate) $q_1 = ?$ (gappy / indeterminate) $q_2 = ?$ (gappy / indeterminate)

$\underbrace{\hspace{10em}}_{q_0 = q_1 = q_2 \text{ (jointly pinned \cdot three-way correlation)}}$

Two branches: DR + EQ (with EQ_y, EQ_g realisations)

{ DR, EQ }



$istate(R_1, R_2) = |(R_1, R_2)\rangle = (1/\sqrt{2}) |DR\rangle + (1/\sqrt{2}) |EQ\rangle$. Two of the eight RCC5⁺ basis vectors carry non-zero coefficient
 $\rightarrow$ $icap = 2 \text{ bits} < 3 \text{ bits}$.

Synthetic-data validation: fois-c / fois-d

Related Born-rule observation studies test the framework's internal consistency.

fois-c	skind classification	fois-d	interrogation strategy
setup 1,800 synthetic systems × Born-rule observation sequences across all 120 ikinds ($n = 3$ predicates). Neural classifier trained to recover the type from observation traces. headline results skind recoverable (77–86%); ikind not ($< 15\%$). → <i>the classical–quantum boundary at the skind level is operational, not just algebraic.</i>		setup Same 1,800 systems × seven interrogation strategies under Hamiltonian dynamics ($T \in \{30, 100\}$, 5 seeds). Tests which observation strategy minimises disturbance. headline results diagonal-first strategy dominates at $T = 100$ (CE 0.971 vs greedy 1.014). Population reaches the predicted skind-level ceiling. → <i>strategic interrogation respects the H_Z structure;</i>	

How these results fit the primer's claim

These studies validate the framework's **internal consistency** — the type-hierarchy is recoverable from its own observation model. The **population-relational frequency-to-amplitude inference** of slide 31 is a closely related but distinct claim that warrants its own experimental treatment.